

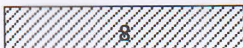


SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2023-October 2023		
Maximum Marks: 30	Examination: In-Semester Examination	Duration: 1 hr 15 min
Programme code: 01 Programme:	Class: FY/SY/TY/LY /MTECH	Semester: I/II/III/IV/V/VI/VII/VIII (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP/ETRX/EXTC/IT/MECH
Course Code: 116U06O511	Name of the Course: Mobile application Development – Flutter	

Question No.		Max. Marks
Q1	Explain following terminologies in DART with the help of an example (Any TWO): i. DART Null safety and null safe operators. ii. Getter and Setter Methods iii. Mixin in DART	10 M
Q2	A. Create an abstract class Employee with private attributes like firstName, lastName, age, empID and methods like calculateSalary() to calculate and display salary of an employee. Then create classes like FullTimeEmployee, PartTimeEmployee, etc., that extend the abstract class and use mixins for additional features like benefits, & overtime. OR B. Create a Calculator class with input taken from user. i. Display welcome and Exit messages. ii. Implement basic Arithmetic operations. iii. Provide Error Handling for following scenarios: • Divide by Zero • Invalid operator Input iv. Create and use Custom Exception for any particular scenario.	10 M
Q3	Fill in the blanks with proper widgets in the given partial flutter code with reference to the given Flutter App GUI: <pre>class BookMenu extends StatefulWidget { const BookMenu({super.key}); @override State<BookMenu> createState() => _BookMenuState(); } class _BookMenuState extends State<BookMenu> { @override Widget build(BuildContext context) { return Scaffold(drawer: SizedBox(width: 270,</pre>	10 M


```

child:  (elevation: 16,
child: Column( mainAxisAlignment: MainAxisAlignment.max,
children: [ Padding( padding:
const EdgeInsetsDirectional.fromSTEB(0, 15, 0, 0),
child:  (width: 250,height: 250,
clipBehavior: Clip.antiAlias,
decoration: const  ( shape: BoxShape.circle,)),
child:  ('https://images...',
fit: BoxFit.cover,)),),
const Text('I_am_Bookworm', style: TextStyle(
color: Color(0x87000000),
fontWeight: FontWeight.w700,
fontSize: 24,)),),
const Text('iambookworm@books.com', style: TextStyle(
color: Color(0x87000000),fontWeight: FontWeight.w500,
fontStyle: FontStyle.italic,)),),
const Divider(thickness: 2, color: Colors.purple,)),
Padding(padding: const EdgeInsetsDirectional.fromSTEB(0, 10, 0, 0),
child:  ( padding:  .zero, shrinkWrap: true,
scrollDirection: Axis.vertical,
children: [ const ListTile(leading: Icon( Icons.person_rounded,)),
title: Text('Personal Info', style: TextStyle( fontSize: 20,)),),
tileColor: Color(0xFFA652D6),)),
const ListTile( leading: Icon( Icons.book_online_outlined,)),
title: Text('My Bookshelf', style: TextStyle(fontSize: 20,)),),
tileColor: Color(0xFFA652D6),)),
ListTile( onTap: () {
 (context,  (builder:
(context) => const MyWishList(),),),)),
leading: const Icon( Icons.bookmark_border_outlined,)),
title: const Text('My Wishlist', style: TextStyle(fontSize: 20,)),),
tileColor: const Color(0xFFA652D6), splashColor: Colors.transparent,)),
const ListTile(leading: Icon(Icons.shopping_cart_outlined,)),
title: Text('My Cart', style: TextStyle(fontSize: 20,)),),

```

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tileColor: Color(0xFFA652D6),),
ListTile( splashColor: Colors.yellow,
onTap: () { 7 (context, 8 (
    builder: (context) => const MyHomePage());},
leading: const Icon(Icons.logout_outlined,),
title: const Text('Logout', style: TextStyle( fontSize: 20,)),),
tileColor: const Color(0xFFA652D6),),),),),),),
appBar: AppBar(
    backgroundColor: const Color(0xFFA652D6),
    title: const Text("Book Catalouge"),),),)}}
class MyWishList extends 9 {
    const MyWishList({super.key});
    @override
    Widget build(BuildContext context) {
        return Scaffold(
            appBar: AppBar(
                title: const Text("My WishList"),
                10: const Color(0xFFA652D6),
            ),
            body: const Center(
                child: Text("My Wish List "),
            ),
        );
    }
}

```

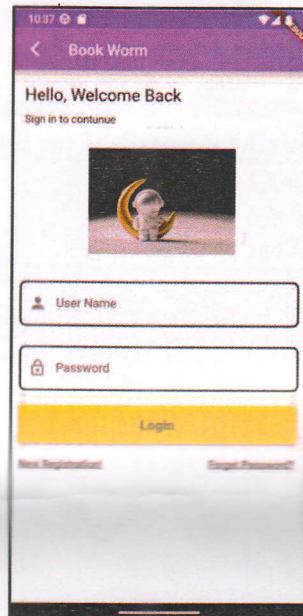
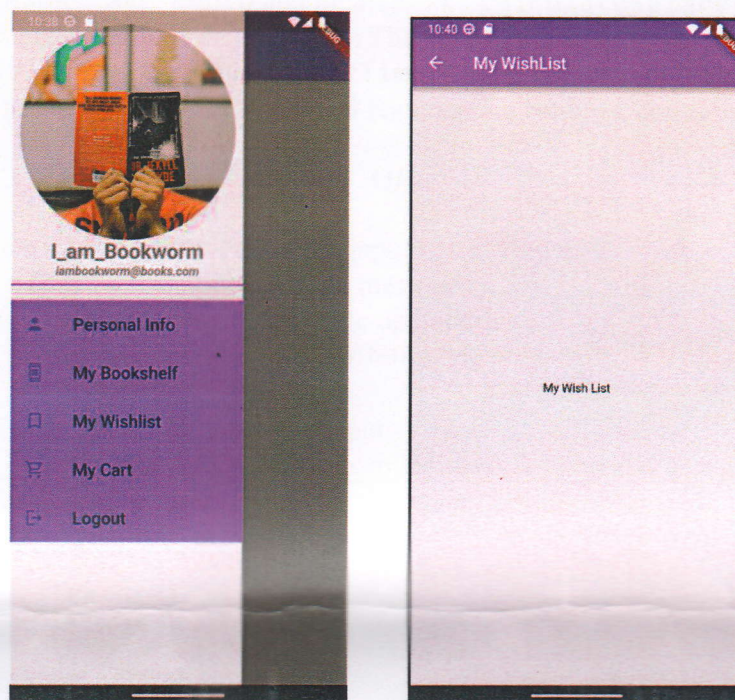



Fig 1: Login Screen



Screen 2: Drawer & My Wishlist Screen